

Total No. of Questions : 8]

SEAT No. :

PB-2497

[Total No. of Pages : 3

[6263]-383

B.E. (Artificial Intelligence and Data Science)

INDUSTRIAL INTERNET OF THINGS

(2019 Pattern) (Semester - VII) (Elective - III) (417523(B))

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve questions Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q. 6, Q.7 or Q. 8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Define Industrial Internet of Things (IIoT). List and briefly explain the components of IIoT architecture. **[8]**
- b) Draw and explain the Reference Architecture of IIoT. **[5]**
- c) Explain the integration of Wireless Sensor Networks (WSN) into the IIoT architecture. **[5]**

OR

- Q2)** a) Discuss the Industrial Internet Architecture Framework (IIAF). Explain its purpose, key principles and how it guides the design and implementation of IIoT systems. **[10]**
- b) Discuss the layers of Industrial IoT (IIoT) architecture. Describe the functionalities and interactions within each layer, focusing on (any 3): **[8]**
- i) IIoT Sensing
 - ii) IIoT Processing
 - iii) IIoT Communication
 - iv) IIoT Networking

- Q3)** a) Compare and contrast the following IIoT cloud platforms w.r.t their features, capabilities and suitability for different industrial applications: **[8]**
- i) Cloud of Things (COT) platforms
 - ii) Predix
 - iii) PTC ThingWorx
 - iv) Microsoft Azure

P.T.O.

- b) Describe various data visualization techniques commonly used in Industrial IoT (IIoT) applications. Explain how these techniques help in representing complex data sets visually for better understanding and analysis. [6]
- c) Differentiate between Software as a Service (SaaS), Platform as a Service (PaaS) and Infrastructure as a Service (IaaS), providing examples for each. [3]

OR

- Q4)** a) Discuss the role of Data Analytics in optimizing Industrial IoT (IIoT) systems. Explain how data analytics techniques can extract valuable insights from IIoT-generated data to improve efficiency, predictive maintenance and decision-making processes. [10]
- b) Explain the concept of Digital Twin in the context of Industrial IoT (IIoT). Discuss the need for Digital Twin technology and its benefits in industrial settings. [7]
- Q5)** a) Explain the importance of security in Industrial IoT (IIoT) deployments. Discuss the potential consequences of security breaches in IIoT systems and their impact on industrial operations. [10]
- b) Discuss the management aspects of cybersecurity in Industrial IoT (IIoT) environments. Explain the roles and responsibilities of stakeholders in managing IIoT security risks and implementing effective cybersecurity policies and procedures. [7]

OR

- Q6)** a) Explain the concept of access control in Industrial IoT (IIoT) environments. Discuss the mechanisms and techniques used to enforce access control policies and permissions in IIoT systems. [10]
- b) Discuss the importance of identity establishment in IIoT security. Explain the methods and protocols used to establish and manage identities for IIoT devices and users. [7]
- Q7)** a) Explain Smart Logistics and its impact on supply chain management. [6]
- b) Describe Smart Irrigation and its benefits in agricultural practices. [6]
- c) Discuss the characteristics and design principles of Industry 4.0. [6]

OR

- Q8)** a) Define Cyber Manufacturing Systems and discuss their importance in modern manufacturing. [6]
- b) Explain the role of IoT in the Healthcare Service Industry and provide examples of IoT-enabled healthcare solutions. [6]
- c) Introduce the concept of Industry 5.0 (Society 5.0) and discuss its potential impact on society. [6]

